Mock Exam – Web Development (50 Marks)

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1. HTML5 Semantic Elements (10 marks)

a) What are semantic elements in HTML5? Explain why they are important for web development. Provide at least 3 examples. (5 marks)

Answer:

Semantic elements in HTML5 are elements that clearly describe their meaning and purpose both to the browser and to developers. Unlike non-semantic elements such as <div> and <span>, semantic elements indicate the type of content they contain.

Importance of semantic elements:

- Improve code readability and maintainability
- Enhance accessibility for screen readers and assistive technologies
- Help search engines understand page structure, improving SEO
- Provide a clear structure for modern web applications

Examples of semantic elements:

- <header> – Represents the introductory content or navigation
- <main> – Contains the main content of the document
- <footer> – Represents footer information such as copyright
- <section> – Groups related content
- <article> – Represents independent, self-contained content

b) Write a simple HTML5 structure using at least 3 semantic elements to define the header, main content, and footer of a web page. (5 marks)

Answer:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">
```

```html
    <title>Semantic HTML Example</title>
</head>
<body>

    <header>
        <h1>My Website</h1>
        <nav>
            <a href="#">Home</a> |
            <a href="#">About</a>
        </nav>
    </header>

    <main>
        <section>
            <h2>Welcome</h2>
            <p>This is the main content of the webpage.</p>
        </section>
    </main>

    <footer>
        <p>&copy; 2025 My Website</p>
    </footer>

</body>
</html>
```

2. Responsive Design (10 marks)

a) Explain the concept of responsive design. Why is it essential in modern web development? (4 marks)

Answer:

Responsive design is an approach to web development where a website automatically adapts its layout and content to different screen sizes and devices such as desktops, tablets, and mobile phones.

Why it is essential:

- Users access websites from various devices
- Improves user experience
- Reduces the need for multiple versions of a website
- Improves SEO ranking
- Ensures accessibility across devices

b) Name and describe two elements/attributes that are important for building responsive websites. (4 marks)

Answer:

1. Viewport Meta Tag
    - Controls how the website is displayed on mobile devices
    - Ensures proper scaling

<meta name="viewport" content="width=device-width, initial-scale=1.0">

2. Flexible Units (%, vw, vh, rem)
    - Allow elements to resize relative to screen size
    - Make layouts flexible instead of fixed

c) What is a media query in CSS, and how is it used for responsive design? Provide a basic example. (2 marks)

Answer:

A media query allows CSS styles to be applied based on device characteristics such as screen width or resolution.

Example:

```
@media (max-width: 768px) {
   body {
      background-color: lightgray;
   }
}
```

This applies styles only when the screen width is 768px or smaller.

3. Flexbox Directions (10 marks)

a) In Flexbox, what are the flex-direction, wrap, order, basis properties? Explain how each affects the layout. (5 marks)

Answer:

- flex-direction: Defines the direction of flex items (row, column, row-reverse, column-reverse)
- flex-wrap: Determines whether items wrap to a new line (nowrap, wrap)
- order: Controls the order of items without changing HTML structure
- flex-basis: Defines the initial size of a flex item before space distribution

Each property helps control alignment, flow, and sizing of flex items.

b) Write CSS to display flex items in a column layout and align them in the center both horizontally and vertically. (5 marks)

Answer:

```
.container {
  display: flex;
  flex-direction: column;
  justify-content: center; /* Vertical alignment */
  align-items: center;     /* Horizontal alignment */
  height: 100vh;
}
```

4. Specificity Rules (10 marks)

a) Explain how CSS specificity works. Provide an example. (5 marks)

Answer:

CSS specificity determines which style rule is applied when multiple rules target the same element. The hierarchy is:

1. Inline styles
2. IDs
3. Classes / attributes / pseudo-classes
4. Elements / pseudo-elements

Example:

```
p {
  color: blue;
}

#text {
  color: red;
```

}

<p id="text">Hello</p>

The text will be red because ID selectors have higher specificity.

b) Calculate the specificity and identify the highest one. (5 marks)

Selector	**Specificity**
.header .nav-item	**0-0-2-0**
#main-content .article h1	**0-1-1-1**
div p	**0-0-0-2**

Highest specificity:
👉 **#main-content .article h1 (contains an ID)**

5. Simple Gradients (10 marks)

a) What is a gradient in CSS? Difference between linear and radial gradients. (4 marks)

Answer:

A gradient is a smooth transition between two or more colors in CSS.

- Linear Gradient: Colors transition in a straight line
- Radial Gradient: Colors radiate from a central point outward

b) Write a CSS rule for a linear gradient from blue to green. (3 marks)

Answer:

background: linear-gradient(blue, green);

c) How can you control the direction of a linear gradient? Modify (b) to create a diagonal gradient. (3 marks)

Answer:

You can control direction using angles or keywords.

background: linear-gradient(45deg, blue, green);

This creates a diagonal gradient from top-left to bottom-right.